

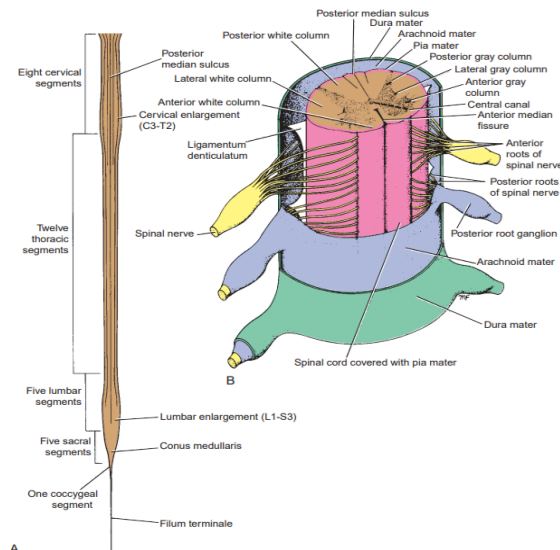
LECTURE 9

THE SPINAL CORD AND THE ASCENDING AND DESCENDING TRACTS

Sumalpong, Princess Kaeleen Loisse S. OPHT 1-1

GROSS APPEARANCE AND STRUCTURE

- **Cylindrical** in shape; continuous with the **medulla oblongata** as it exits the **foramen magnum**, and **terminates at the lower border of L1**.
- Occupies **2/3 of the vertebral canal**, is surrounded by **three meninges (dura, arachnoid, and pia mater)**, and is **nourished by cerebrospinal fluid (CSF)**.
- Tapers off into the **conus medullaris**.
- The **filum terminale** (a prolongation of the pia mater) descends **to attach to the posterior surface of the coccyx**.
- Possesses a **shallow anterior median sulcus** and a **posterior median sulcus**.
- **Spinal Nerves – 31 pairs.**
 - **Anterior roots** - efferent fibers
 - **Posterior roots** - afferent fibers
 - **Posterior root ganglion** - cells that give rise to peripheral and central nerve fibers.



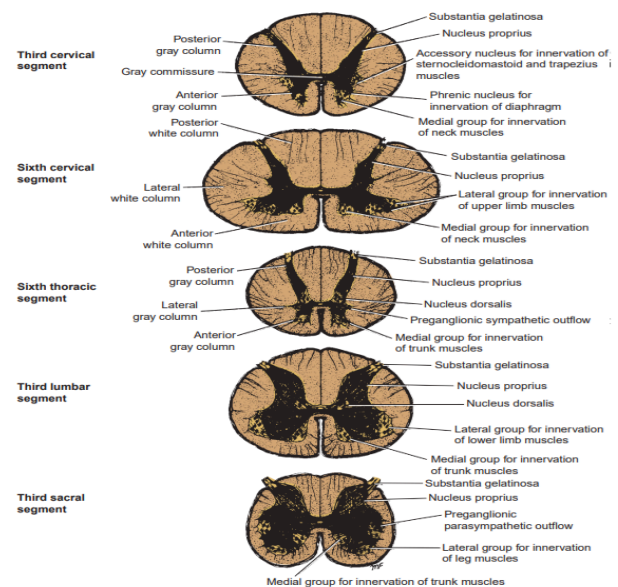
STRUCTURE OF THE SPINAL CORD

GRAY MATTER

- **H-shaped pillar** with anterior and posterior gray columns (horns) **united by a thin gray commissure containing the central canal**.
- The **lateral gray column** is restricted to the **thoracic and upper lumbar segments**.

Anterior Gray Column Groups

- Axons exit via the anterior roots as alpha efferents to **innervate skeletal muscle**
 - a. As smaller gamma efferents to **innervate intrafusal fibers of neuromuscular spindles**.
- **3 basic groups:**
 1. **Medial Group** – Present in most segments
 - Innervates **skeletal muscles of the neck and trunk** (including intercostal and abdominal musculature).
 2. **Central Group** – **Smallest** group, present in some cervical and lumbosacral segments.
 - **Accessory Nucleus** - SCM and trapezius
 - **Lumbosacral Nucleus** (L2–L5, unknown distribution).
 3. **Lateral Group** – Present in cervical and lumbosacral segments
 - Innervates **skeletal muscles of the limbs**.



Posterior Gray Column Groups

- **Substantia Gelatinosa** – At the apex throughout the length of the cord.
 - Composed of **Golgi Type II neurons**
 - Receives afferent fibers for **pain, temperature, and touch**, as well as **descending**

supraspinal fibers.

- **Nucleus Proprius** - Associated with **proprioception (position/movement), two-point discrimination, and vibration.**
- **Nucleus Dorsalis (Clarke's Column)** – Associated with **proprioceptive endings (neuromuscular and tendon spindles).**
- **Visceral Afferent Nucleus** - Receives **visceral afferent information.**

Lateral Gray Column Group

- **Intermediolateral Group**
 - T1 to L3 cells - preganglionic sympathetic fibers
 - S2 to S4 cells - preganglionic parasympathetic fibers.

WHITE MATER

- Surrounds the gray matter and is divided into **anterior, lateral, and posterior columns.**
- Its white appearance is due to a **high proportion of myelinated nerve fibers.**

THE ASCENDING TRACTS OF THE SPINAL CORD

- Conduct **afferent information** (may or may not reach consciousness).
- **2 groups:**
 1. **Exteroceptive** - from *outside* the body
 2. **Proprioceptive** - from *inside* the body

General Anatomical Organization

- **1st-order neuron** - resides in the posterior root ganglion.
- **2nd-order neuron** - gives rise to an axon that decussates (crosses over) and ascends to synapse with the 3rd-order neuron (usually in the thalamus.)
- **3rd-order neuron** - projects to the sensory region of the cerebral cortex.

Functions of the Ascending Tracts

1. **LATERAL SPINOTHALAMIC TRACT** - pain and temperature sensory pathway.

- **1st-order fibers** - travel 1–2 segments in the posterolateral tract of Lissauer, then synapse.
- **2nd-order axons** - cross obliquely in the anterior gray/white commissures within 1 segment, ascend as the lateral spinothalamic tract in the spinal lemniscus, and synapse in the thalamus.
- **3rd-order fibers** - project via the internal capsule and corona radiata to the postcentral gyrus.

2. **ANTERIOR SPINOTHALAMIC TRACT** - light (crude) touch and pressure pathways.

- **1st-order fibers** - travel in the tract of Lissauer and synapse in the substantia gelatinosa.
- **2nd-order axons** - cross very obliquely in the anterior commissures over several segments, ascend in the spinal lemniscus, and synapse in the ventral posterolateral (VPL) nucleus of the thalamus.
- **3rd-order fibers** - project to the postcentral gyrus.

3. **POSTERIOR WHITE COLUMN: FASCICULUS GRACILIS AND FASCICULUS CUNEATUS** - Discriminative touch, vibratory sense, and conscious muscle joint sense.

- **1st-order fibers** - enter the ipsilateral posterior column, ascend to synapse in the nucleus gracilis and nucleus cuneatus of the medulla.
- **2nd-order axons (internal arcuate fibers)** - decussate, ascend as the medial lemniscus, and synapse in the VPL nucleus of the thalamus.
- **3rd-order fibers** - project to the postcentral gyrus.

- **Cuneocerebellar Tract** - Conveys muscle joint sense to the cerebellum.
 - Also known as **posterior external arcuate fibers**
 - **2nd-order axons** from the nucleus cuneatus travel as **posterior external arcuate fibers** to enter the ipsilateral cerebellum via the inferior cerebellar peduncle.

4. **POSTERIOR SPINOCEREBELLAR TRACT** - Unconscious muscle joint sense (trunk/lower limbs)
- Controls **posture and coordinates movements**.
 - **1st-order fibers** - synapse with 2nd-order neurons in the nucleus dorsalis (Clarke's column).
 - **2nd-order axons** - ascend ipsilaterally and enter the cerebellum through the inferior cerebellar peduncle.

5. **ANTERIOR SPINOCEREBELLAR TRACT** - Conveys muscle joint information from the muscle spindles, tendon organs, and trunk/limb receptors.
- Skin and superficial fascia information
 - **1st-order fibers** synapse in the nucleus dorsalis.
 - The **majority of 2nd-order axons decussate and ascend in the contralateral white column**, while a **minority ascend ipsilaterally**.

6. **CUNEOCEREBELLAR TRACT** - Conveys muscle joint sense to the cerebellum.

7. **SPINOTECTAL TRACT** - Afferent pathway for spinovisual reflexes (eyes/head movement toward stimulus).
- **2nd-order neurons** decussate and ascend in the anterolateral white column to terminate in the superior colliculus of the midbrain.

8. **SPINORETICULAR TRACT** - Influences levels of consciousness.
- **2nd-order axons** ascend ipsilaterally in the lateral white column mixed with the lateral spinothalamic tract to **terminate in the reticular formation of the medulla, pons, and midbrain**.

9. **SPINO-OLIVARY TRACT** - conveys cutaneous and proprioceptive organ information to the cerebellum.
- **2nd-order axons** decussate, ascend at the junction of the anterior and lateral columns, and **synapse in the inferior olivary nuclei of the medulla**.
 - From there, axons decussate again to enter the cerebellum via the inferior cerebellar peduncle

THE DESCENDING TRACTS OF THE SPINAL CORD

- Also known as **Lower Motor Neurons**
- Motor neurons in the anterior gray columns **send axons to innervate muscle through anterior roots of spinal nerves**.
- **Descending Tracts** - nerve bundles segregated from supraspinal nerve centers.
- **Upper Motor Neurons** - provide pathways that influence motor activities

General Anatomical Organization

- **1st-order neuron** - cerebral cortex
- **2nd-order neuron** - internuncial neurons in the anterior gray column
- **3rd-order neuron** - lower motor neuron (LMN) in the anterior gray column, which **innervates skeletal muscle**.

Functions of the Descending Tracts

1. **CORTICOSPINAL TRACT** - Rapid, skilled, voluntary movements (especially distal limb portions).

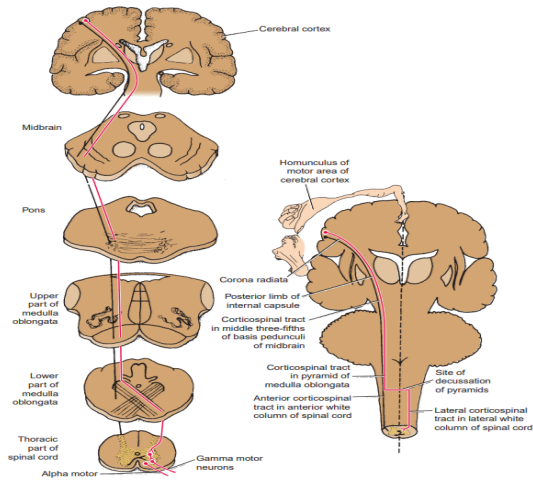


Figure 4-21 Corticospinal tracts.

2. RETICULOSPINAL TRACT - Facilitates/inhibits alpha and gamma motor neuron activity.

- Pathway for hypothalamus to control autonomic outflow.

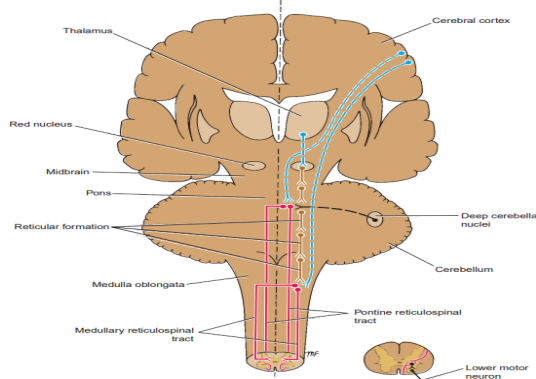
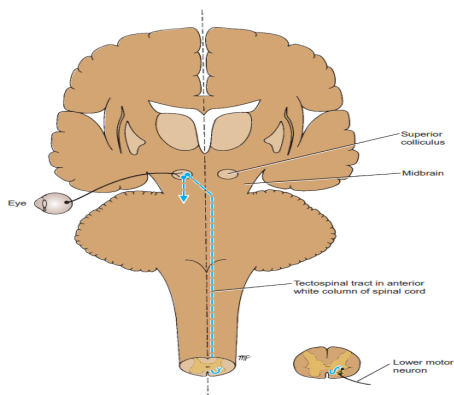


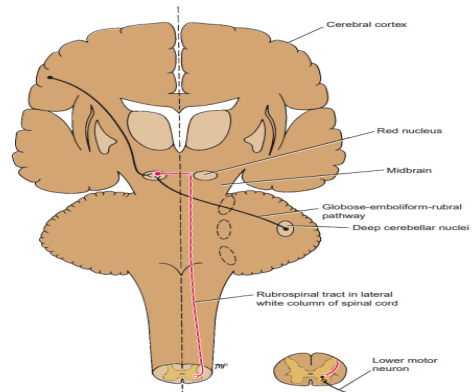
Figure 4-22 Reticulospinal tracts.

3. TECTOSPINAL TRACT - reflex postural movements in response to visual stimuli.



4. RUBROSPINAL TRACT - Facilitates flexor muscles and inhibits

extensor/antigravity muscles via alpha and gamma neurons.



5. VESTIBULOSPINAL TRACT - Facilitates extensor muscles, inhibits flexor muscles; maintains posture and balance.

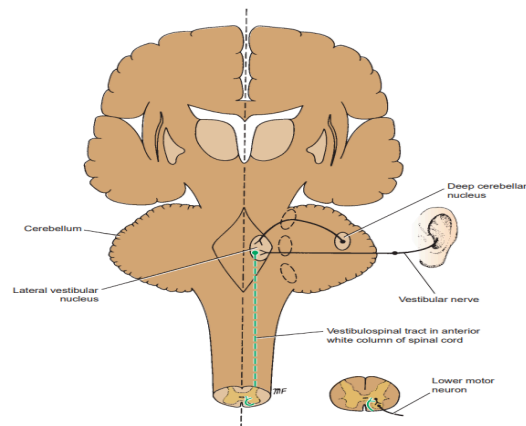


Figure 4-25 Vestibulospinal tract.

6. OLIVOSPINAL TRACT - Associated with muscular activity.

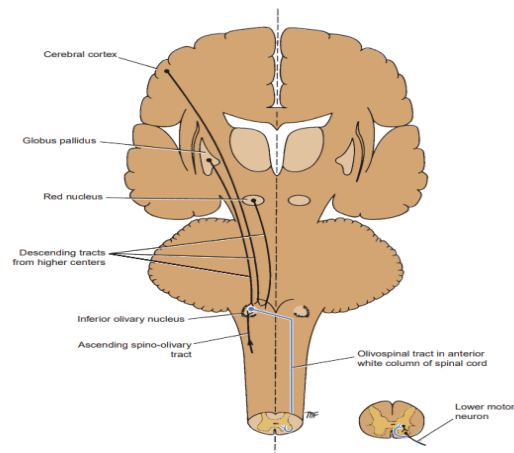


Figure 4-26 Olivospinal tract. There is now considerable doubt as to the existence of this tract as a separate pathway.

7. DESCENDING AUTONOMIC FIBERS - Control of visceral activity.

Pathways:

CORTICOSPINAL	$\frac{1}{3}$ primary motor cortex; $\frac{1}{3}$ secondary cortex; $\frac{1}{3}$ parietal lobe $\frac{2}{3}$ precentral gyrus; $\frac{1}{3}$ post central gyrus Arises from pyramidal cells in the 5th layer of the cerebral cortex. Runs through the corona radiata > posterior limb of internal capsule > middle $\frac{3}{5}$ of basis pedunculi > pons > medullary pyramids. Decussation: The <u>majority</u> cross at the <u>medullary-spinal junction</u> to form the Lateral Corticospinal Tract (LCST) in the lateral column. - A <u>minority</u> descend ipsilaterally as the Anterior Corticospinal Tract, crossing at their destination segment in the cervical/upper thoracic regions.
RETICULOSPINAL	Pontine Reticulospinal Tract - Arises in the pons and descends uncrossed in the anterior white column. Medullary Reticulospinal Tract - Arises in the medulla and descends crossed and uncrossed in the lateral white column.
TECTOSPINAL	Arises in the superior colliculus of the midbrain Decussates near the medial longitudinal fasciculus

	Descends in the anterior white column to terminate in the upper cervical segments.
RUBROSPINAL	Arises in the red nucleus of the midbrain - Decussates immediately at the midline of the red nucleus - Descends through the pons and medulla into the lateral white column to reach the anterior gray column.
VESTIBULOSPINAL	Arises in the vestibular nuclei - Descends uncrossed through the medulla into the anterior white column Synapses on internuncial neurons of the anterior gray column.
OLIVOSPINAL	Arises in the inferior olivary nucleus Descends in the lateral white column to influence anterior gray column motor neurons. (Note: There is significant doubt regarding its existence as a separate pathway).

Pathway	Function	Origin	Site of Crossover	Destination	Branches to
Corticospinal tracts	Rapid, skilled, voluntary movements, especially distal ends of limbs	Primary motor cortex (area 4), secondary motor cortex (area 6), parietal lobe (areas 3, 1, and 2)	Most cross at decussation of pyramids and descend as lateral corticospinal tracts; some continue as anterior corticospinal tracts and cross over at level of destination	Internuncial neurons or alpha motor neurons	Cerebral cortex, basal nuclei, red nucleus, olivary nuclei, reticular formation
Reticulospinal tracts	Inhibit or facilitate voluntary movement; hypothalamus controls sympathetic, para-sympathetic outflows	Reticular formation	Some cross at various levels	Alpha and gamma motor neurons	Multiple branches as they descend
Tectospinal tract	Reflex postural movements concerning sight	Superior colliculus	Soon after origin	Alpha and gamma motor neurons	?
Rubrospinal tract	Facilitates activity of flexor muscles and inhibits activity of extensor muscles	Red nucleus	Immediately	Alpha and gamma motor neurons	?
Vestibulo-spinal tract	Facilitates activity of extensor inhibits flexor muscles	Vestibular nuclei	Uncrossed	Alpha and gamma motor neurons	?
Olivospinal tract	??	Inferior olivary nuclei	Cross in brainstem	? Alpha and gamma motor neurons	—
Descending autonomic fibers	Control sympathetic and para-sympathetic systems	Cerebral cortex, hypothalamus, amygdaloid complex, reticular formation		Sympathetic and parasympathetic outflows	—

Corticospinal Somatotopic Organization

Note:

- **Internal Capsule** – Cervical fibers lie close to the genu; lower extremity fibers lie posteriorly.
- **Basis Pedunculi** – Cervical fibers are located medially; lower extremity fibers are located laterally.

ASCENDING TRACTS INJURIES

- **Lateral Spinothalamic Tract Lesion**
 - **Contralateral loss of pain and thermal sensibilities below the level of the lesion** (inability to perceive a pinprick or hot/cold items).
- **Anterior Spinothalamic Tract Lesion**
 - **Contralateral loss of light touch and pressure sensibilities below the level of the lesion** (cannot perceive cotton wisps or blunt object pressure).
- **Posterior White Column Lesion**
 - **Ipsilateral loss of vibration sense, loss of tactile discrimination, and loss of conscious muscle joint sense** (inability to perceive position/movements of limbs) **below the injury level**.
 - **Light touch** remains largely unaffected.

Somatic vs. Visceral Pain Differentiation

- **Somatic Pain** – Originates **from skin, muscles, joints, bones, or connective tissue**.
 - Supplied by **somatic nerves**.
 - Quality is **sharp, well-defined, localized, easy to pinpoint, and worsens with movement/posture changes**.
- **Visceral Pain** – Originates **from internal organs**.
 - Supplied by **autonomic nerves**.
 - Quality is **dull, deep, cramping, aching, and poorly localized**; often associated

with symptoms like nausea, vomiting, sweating, or blood pressure changes.

DESCENDING TRACTS INJURIES

- **Upper Motor Neuron (UMN) Lesion / Corticospinal Tract Lesion**
 - **Babinski Reflex** – Response to scratching the **lateral sole** changes from normal plantar flexion to **dorsiflexion of the great toe and outward fanning of the other toes**.
 - **Superficial Abdominal Reflex** – Muscles **fail to contract** when abdominal skin is **scratched**.
 - **Cremasteric Reflex** – Scrotum/muscles **fail to contract when the medial thigh skin is stroked**.
 - **Functional Loss** – Loss of performance of fine-skilled voluntary movements, particularly at the **distal end of the limbs**.
- **Extrapyramidal Tract Lesion (Descending Tracts other than Corticospinal)**
 - Severe paralysis with **minimal or no muscle atrophy**.
 - Spasticity/hypertonicity (lower limbs fixed in extension, upper limbs fixed in flexion).
 - Exaggerated **deep muscle reflexes and clonus** (e.g., finger flexors, quadriceps femoris, calf muscles).
 - **Clasp-knife reaction** (initial catch and resistance during passive joint movement, which then suddenly gives way).
- **Lower Motor Neuron (LMN) Lesion**
 - Flaccid paralysis, muscle atrophy, and **complete loss of reflexes in the supplied muscles**.
 - **Muscular Fasciculation** – Brief twitching visible under the skin,

- occurring specifically during the slow destruction of the LMN cell.
- **Muscular Contracture** – **Permanent shortening of paralyzed muscles**, often presenting in the un-opposed antagonist muscles.
- **Reaction of degeneration.**

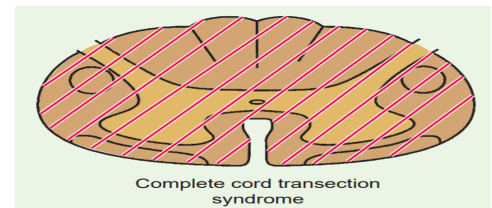
Types of Paralysis

- **Hemiplegia** – Paralysis of one side of the body (upper limb, trunk, and lower limb).
- **Monoplegia** – Paralysis of one limb only.
- **Diplegia** – Paralysis of two corresponding limbs (arms or legs).
- **Paraplegia** – Paralysis of both lower limbs.
- **Quadriplegia** – Paralysis of all four limbs.

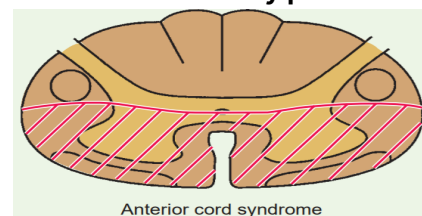
CLINICAL SYNDROMES AFFECTING THE SPINAL CORD

- **Spinal Shock Syndrome**
 - Occurs immediately after acute, severe spinal cord damage.
 - All cord functions below the injury level become depressed or lost, leading to **temporary sensory impairment, flaccid paralysis, and depressed reflexes**.
 - It persists for less than **24 hours**.
 - As it diminishes, neurons regain excitability, and the classic signs of UMN loss emerge.
- **Complete Cord Transection Syndrome**
 - Caused by fracture-dislocations, **bullet/stab wounds, or expanding tumors**.
 - Results in **complete loss of all sensibility and voluntary movement** below the lesion.
 - *Clinical Features*

- **Bilateral LMN paralysis** and muscular atrophy at the segment level of the lesion.
- **Bilateral spastic paralysis and bilateral loss** of all sensations *below* the lesion level.
- **Bladder and bowel control are entirely lost.**



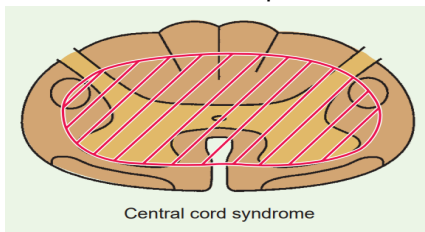
- **Anterior Cord Syndrome**
 - Caused by vertebral fractures, anterior spinal artery occlusion, or herniated intervertebral discs.
 - *Clinical Features*
 - **Bilateral LMN paralysis** at the lesion level.
 - **Bilateral spastic paralysis** *below* the lesion level.
 - **Bilateral loss of pain, temperature, and light touch sensations** *below* the lesion level.
 - Tactile discrimination, vibratory sense, and proprioception are **fully preserved**.



- **Central Cord Syndrome**
 - Caused by **hyperextension injuries of the cervical spine**.
 - *Clinical Features*
 - **Bilateral LMN paralysis** at the lesion

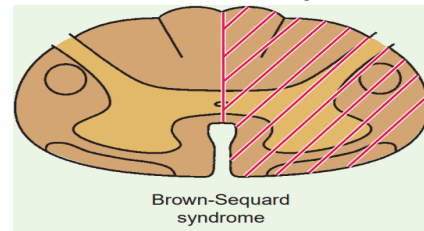
level.

- **Bilateral spastic paralysis** *below* the lesion level.
- **Bilateral loss of pain, temperature, light touch, and pressure sensations** *below* the lesion level.
- Crucially displays characteristic **sacral sparing** (sensory and motor functions of the sacral dermatomes are preserved).



- **Brown-Séquard Syndrome (Hemisection of the Cord)**
 - Caused by fracture-dislocations, stab/bullet wounds, or expanding tumors.
 - *Clinical Features*
 - **Ipsilateral LMN paralysis** *at* the lesion segment, accompanied by an **ipsilateral band of cutaneous anesthesia**.
 - **Ipsilateral spastic paralysis** *below* the level of the lesion.
 - **Ipsilateral loss of tactile discrimination, vibratory, and proprioceptive senses** *below* the lesion level.
 - **Contralateral loss of pain and temperature sensations** *below* the lesion level.
 - Contralateral loss of tactile sensation is present but

incomplete.



DISEASE-SPECIFIC SYNDROMES

- **Poliomyelitis** – A viral infection targeted specifically at the **anterior gray columns**.
 - Causes LMN-type paralysis and severe muscle wasting (**lower extremities are affected more than upper extremities**); can threaten respiration if cervical segments are involved.
- **Multiple Sclerosis** – An autoimmune, hereditary, or infectious condition causing **demyelination of both ascending and descending tracts**.
 - Breakdown of **axonal insulation** reduces action potential velocity and can ultimately block signal conduction.
 - Most common symptoms are **limb weakness and ataxia**.
- **Amyotrophic Lateral Sclerosis (ALS / Lou Gehrig Disease)** – A **progressive disease** strictly confined to the corticospinal tracts and the motor neurons of the anterior gray columns.
 - Combines **LMN signs** (progressive muscular atrophy, paresis, fasciculations) and **UMN signs** (spasticity, Babinski sign).

LECTURE 10

THE AUTONOMIC NERVOUS SYSTEM

Sumalpong, Princess Kaeleen Loisse S. OPHT 1-1

ORGANIZATION OF THE AUTONOMIC NERVOUS SYSTEM (ANS)

- Sympathetic Division
- Parasympathetic Division

SYMPATHETIC PART OF THE AUTONOMIC SYSTEM

- **Larger** of the two ANS components.
- Widely distributed throughout the body, including the following structures:
 - Heart and lungs
 - Smooth muscle in the walls of many blood vessels
 - Hair follicles; Sweat glands
 - Many abdominopelvic viscera

Functional Role

- To prepare the body for an emergency ("**fight-or-flight**" response).
- **Tachycardia** – Increased heart rate.
- **Blood Pressure Elevation** – overall increase in blood pressure.
- **Selective Vasomotor Changes** – Vasoconstriction in the skin and intestines, paired with vasodilation of the arterioles supplying skeletal muscles.
- **Blood Redistribution** – Blood is shunted away from the skin and GI tract toward vital organs like the brain, heart, and skeletal muscle.
- **Ocular Changes** – Mydriasis (pupil dilation).
- **Respiratory Changes** – Bronchoconstriction.
- **Visceral Sphincter Closure** – Closure of sphincters within the intestine and the bladder wall.
- **Somatic Activation** – Hair follicle **erection** (pilomotor activity) and increased sweating.

Neuronal Outflow Pathway

1. **Primary (Presynaptic/Preganglionic)**

Neuron – Cell bodies are located in the **intermediolateral gray column of the spinal cord** or within specific brainstem nuclei.

- a. They give rise to small, myelinated B fibers.

2. **Secondary (Postsynaptic/Postganglionic)**

Neuron – Located within an autonomic ganglion, **sending its unmyelinated axon directly to the target organ**.

Sympathetic Efferent Fiber Systems

1. Preganglionic Efferent System

- **Thoracolumbar Outflow** – Arises from spinal cord segments T1 to L2 or L3.
- **White Communicating Rami** – Formed by mostly myelinated preganglionic fibers exiting the thoracic and lumbar spinal nerves to enter the sympathetic trunks or chains.
- **Sympathetic Trunks** – Located on the lateral sides of the thoracic and lumbar vertebral bodies.
- **Splanchnic Nerve Pathways** – Preganglionic fibers tracking through lower splanchnic nerves (from T7 and below) bypass the trunk ganglia to synapse in the Celiac and Superior Mesenteric Ganglia.
 - Fibers traveling via splanchnic nerves from T12 to L1 track to the Inferior Mesenteric Ganglion and the Hypogastric Plexus.

The Adrenal Gland Exception

- Preganglionic sympathetic axons within the splanchnic nerves > adrenal medulla.
- They provide direct synaptic input to **Chromaffin Cells**.

Postganglionic Efferent System

- **Gray Communicating Rami** – Formed by mostly **unmyelinated postganglionic axons**.
- They rejoin each of the spinal nerves **to distribute vasomotor, pilomotor, and sweat gland innervation throughout somatic areas** of the body.
- **Superior Cardiac Nerves > Cardiac Plexus > Myocardium**
 - **Superior Cardiac Nerves** – Arise from the cervical chain, passing into the *Cardiac Plexus* to supply the myocardium.
- **Thoracic Vasomotor Branches** – Fibers from T5 supply the thoracic aorta, tracking through the **Posterior Pulmonary Plexus** to provide dilator fibers to the bronchi.

PARASYMPATHETIC PART OF THE AUTONOMIC SYSTEM

- **Parasympathetic (Craniosacral) Division**
- **LONG preganglionic axons** and **SHORT postganglionic axons**.
- **It originates from two central nervous system zones:**
 - Preganglionic cell bodies reside in the middle three segments of the sacral spinal cord (S2 to S4).
 - *Edinger-Westphal Nucleus*
 - *Superior Salivatory Nuclei*
 - *Inferior Salivatory Nuclei*.

Cranial and Sacral Distribution Pathways

- **Sacral Outflow (S2–S4)** – Axons run without interruption from their central origin directly to either the wall of the viscus they supply or to terminal ganglion cells.

- These terminal cells are embedded in the walls of the intestinal tract within the **Plexus of Meissner (submucosal)** and **Plexus of Auerbach (myenteric)**.
- **Cranial Nerves** – Four specific cranial nerves transport parasympathetic innervation out of the brainstem:
 - **Oculomotor Nerve (CN III)**
 - **Facial Nerve (CN VII)**
 - **Glossopharyngeal Nerve (CN IX)**
 - **Vagus Nerve (CN X)**

AUTONOMIC PLEXUSES

- **Large, interconnected** networks of nerves that serve as **conduits for distributing sympathetic, parasympathetic, and visceral afferent fibers**.
- **Cardiac Plexus**
 - Location: **bifurcation of the trachea** and the **roots of the great vessels at the base of the heart**.
 - Formed by **cardiac sympathetic nerves** and the **cardiac branches of the vagus nerve**
 - Distributes to the **myocardium and emerging vessels**.
- **Pulmonary Plexuses (Left and Right)**
 - Joined with the **cardiac plexus**
 - Location: **primary bronchi** and **pulmonary arteries** at the roots of the lungs.
 - **Formed by the vagus nerve and upper thoracic sympathetic nerves** to supply pulmonary vessels and bronchi.

- **Celiac (Solar) Plexus**

- Location: **epigastric region** surrounding the abdominal aorta.
- Formed by **vagal fibers** (via the esophageal plexus), **sympathetic fibers from the celiac ganglia, and sympathetic fibers descending from the thoracic aortic plexus.**
- Projects to most abdominal viscera via subplexuses:
 - Phrenic, Hepatic, and Splenic plexuses
 - Superior Gastric, Suprarenal, and Renal plexuses
 - Spermatic or Ovarian plexuses
 - Abdominal Aortic, Superior Mesenteric, and Inferior Mesenteric plexuses

- **Hypogastric Plexus**

- Location: **fifth lumbar vertebra** and the **promontory of the sacrum.**
- Receives sympathetic fibers from the aortic plexus and lumbar trunk ganglia, and parasympathetic fibers from the pelvic splanchnic nerve.
- **The pelvic plexus** - lie on the sides of the rectum and project to pelvic organs and genitals via specific subplexuses running along branches of the hypogastric artery:
 - **Middle Hemorrhoidal Plexus** – Rectum
 - **Vesical Plexus** – Bladder, seminal vesicles, and ductus deferens
 - **Prostatic Plexus** –

Prostate, seminal vesicles, and penis

- **Vaginal Plexus** – Vagina and clitoris

- **Uterine Plexus** – Uterus and uterine tubes

AUTONOMIC INNERVATION OF THE HEAD

- **Skin of the Face and Scalp** – Smooth muscle, glands, and vessels receive **postsynaptic sympathetic innervation only**
 - From the **superior cervical ganglion** via the carotid plexus.
- **Deeper Cranial Structures** – Intrinsic eye muscles, salivary glands, and the mucous membranes of the nose and pharynx receive **dual sympathetic and parasympathetic innervation.**

There are **4 pairs of autonomic ganglia** located in the head:

1. **Ciliary Ganglion** – Between the **optic nerve and lateral rectus muscle** in the posterior orbit.
 - **Parasympathetic Root** – Cells in or near the Edinger-Westphal nucleus of CN III.
 - **Sympathetic Root** – Postganglionic fibers from the superior cervical ganglion via the internal carotid plexus.
 - **Sensory Root** – Nasociliary branch of the ophthalmic nerve (CN V1)
 - Travels via **12 short ciliary nerves** to supply the ciliary muscle of the lens and the pupillary constrictor. (*Note: Iris pupillary dilator is supplied directly by the sympathetic root).*
2. **Sphenopalatine (Pterygopalatine)**

Ganglion – Deep in the pterygopalatine fossa, anatomically associated with the **Maxillary Nerve**.

- **Parasympathetic Root** – Cells of the superior salivatory nucleus via the glossopalatine and greater petrosal nerves.
- **Sympathetic Root** – Internal carotid plexus by way of the deep petrosal nerve.
- **Sensory Root** – Maxillary Nerve (CN V2).
- **Target Distribution** –
 - **Pharyngeal rami** – Mucous membranes of the pharyngeal roof.
 - **Nasal/Palatine rami** – Nasal cavity, uvula, tonsils, hard/soft palates.
 - **Orbital rami** – Orbit and lacrimal glands.

3. **Otic Ganglion** – Medial to the mandibular nerve, just below the foramen ovale in the infratemporal fossa.
 - **Parasympathetic Root** – Inferior salivatory nucleus via the tympanic plexus and lesser superficial petrosal nerve.
 - **Sympathetic Root** – Plexus on the middle meningeal artery.
 - **Sensory Root** – CN IX and the geniculate ganglion of CN VII.
 - **Secretory and sensory fibers to the parotid gland.** (*Some motor fibers pass through to supply tensor tympani and tensor veli palatini*).
4. **Submaxillary Ganglion** – Medial side of the mandible **between the lingual nerve and submaxillary duct**.
 - **Parasympathetic Root** – Superior salivatory nucleus of CN VII via the glossopalatine,

chorda tympani, and lingual nerves.

- **Sympathetic Root** – Plexus of the external maxillary artery.
- **Sensory Root** – Geniculate ganglion via the glossopalatine, chorda tympani, and lingual nerves.
- **Secretory fibers to the submaxillary** (submandibular) and sublingual glands.

TABLE 20-1 Pain Innervation of the Viscera.

Division	Nerve(s) or Segment(s)	Structures
Parasympathetic	Vagus	Esophagus, larynx, trachea
Sympathetic	Splanchnic (T7–L1)	Stomach, spleen, small viscera, colon, kidney, ureter, bladder (upper part), uterus (fundus), ovaries, lungs
	Somatic (C7–L1)	Parietal pleura, diaphragm, parietal peritoneum
Parasympathetic	Pelvic (S2–4)	Rectum, trigone of the bladder, prostate, urethra, cervix of the uterus, upper vagina

PATHWAYS TO THE SPINAL CORD

- Afferent fibers enter via the **middle sacral, thoracic, and upper lumbar spinal nerves**.
- They carry **sensations of hunger, nausea, and poorly localized visceral pain**.
 - **Thoracic & Upper Lumbar Nerves** - visceral pain impulses from the heart, upper digestive tract, kidney, and gallbladder.
 - **Sacral Nerves (S2–S4)** - stimuli from pelvic organs regulating reflexes like micturition, defecation, and sexual responses.

PATHWAYS TO THE BRAIN STEM

- **Glossopharyngeal (CN IX) and Vagus (CN X) nerves** - sensory input to the brainstem from the heart, great vessels, respiratory tract, and GI tract.
 - **Baroreceptors** - Via the inferior ganglia of CN IX and X to regulate blood pressure

- and heart rate.
- **Chemoreceptors** - Located in the aortic and carotid bodies; sensitive to hypoxia to adjust respiratory rate and depth.

THE ENTERIC NERVOUS SYSTEM (ENS)

- Complex collection of neurons embedded within the walls of the gastrointestinal tract, often termed the "***intrinsic nervous system of the GI tract***".
- Continuously modulated by **sympathetic and parasympathetic inputs.**
- **Regulates motility, secretory activity, vascular tone, and inflammation.**

Major Enteric Plexuses

- **Meissner's (Submucosal) Plexus** – Most prominent inside the **small intestine.**
 - It primarily regulates local secretory activity and innervates mucosal blood vessels.
- **Auerbach's (Myenteric) Plexus** – Between the inner circular and outer longitudinal smooth muscle layers across the entire length of the GI tract.
 - It primarily regulates peristaltic motility and movement.

Autonomic Modulation of the ENS

- **Parasympathetic Input** – Carried mostly by the **vagus nerves** (supplying the upper GI tract) and **sacral nerves** (modulating lower colon and rectum contractility).
- **Sympathetic Input** – Decreases motility and secretions.
- **Sensory Transmission** – Enteric sensory data passes back to the central nervous system via the vagus

and splanchnic nerves, **utilizing primary afferent neurons located in the nodose ganglia.**

- Most preganglionic parasympathetic neurons are **cholinergic.**
- Most preganglionic sympathetic neurons are **adrenergic.**

TRANSMITTER SUBSTANCES

Principal Neurotransmitters

- **Acetylcholine (ACh)** and **Norepinephrine (NE).**
- **All Preganglionic Neurons** (both sympathetic and parasympathetic) secrete **Acetylcholine.**
 - *Exception:* Preganglionic Fibers to the Adrenal Medulla–Norepinephrine
- **Parasympathetic Postganglionic Neurons** secrete **Acetylcholine.**
- **Sympathetic Postganglionic Neurons** secrete **Norepinephrine.**

RECEPTORS

1. Cholinergic Receptors

- Bind Acetylcholine (ACh) and are utilized across both divisions.

Nicotinic Receptors

- **Type** – Ligand-gated ion channels.
- **Location** – Found on **all autonomic ganglia** (both sympathetic and parasympathetic), **chromaffin cells of the adrenal medulla, and the somatic neuromuscular junction.**
- **Function** – Always excitatory; drives fast ganglionic neurotransmission from preganglionic to postganglionic fibers.

Muscarinic Receptors

- **Type** – G-protein coupled receptors.
- **Location** – Found on all target organs of parasympathetic postganglionic nerves, as well as selected sympathetic postganglionic targets (such as sweat glands).
- **Function** – Can be excitatory or inhibitory depending on the tissue and G-protein subtype.
- **Subtypes:**
 - **M1** – Located in the CNS and gastric glands; excitatory secretions.
 - **M2** – Located in the myocardium; inhibitory low heart rate and conduction velocity.
 - **M3** – Located in smooth muscles and exocrine glands; excitatory (drives bronchoconstriction, miosis, and increased glandular secretion).

2. Adrenergic Receptors

- Bind **Norepinephrine (NE)** and **Epinephrine (Epi)** and are primarily utilized by the **sympathetic division**.

Alpha Receptors

- **a1 Receptors** – Located on smooth muscle and the pupillary radial muscle of the iris.
 - Vasoconstriction, pupil dilation (mydriasis), and increased blood pressure.
- **a2 Receptors** – Located on presynaptic nerve terminals and within the CNS.
 - Negative feedback loop that inhibits further NE release, down-regulating sympathetic outflow.

Beta Receptors

- **B1 Receptors** – Located **at heart, kidneys**

- Increases heart rate, cardiac contractility, conduction velocity, and stimulates renin secretion.

- **B2 Receptors** – Located on bronchi, the gastrointestinal tract, and skeletal muscle blood vessels

- Bronchodilation, vasodilation of skeletal muscle beds, and decreased GI motility.

- **B3 Receptors** – Located in adipose tissue and the detrusor muscle of the bladder.

- Simulates lipolysis and bladder wall relaxation.

Receptor	Location	Neurotransmitter	Effect
Nicotinic	Autonomic ganglia, adrenal medulla	ACh	Excitatory (ganglionic transmission)
Muscarinic M1	CNS, gastric glands	ACh	↑ Secretions
Muscarinic M2	Heart	ACh	↓ HR
Muscarinic M3	Smooth muscles, glands	ACh	Constriction, ↑ secretion
Alpha-1	Vessels, iris	NE/Epi	Vasoconstriction, mydriasis
Alpha-2	Presynaptic terminals	NE/Epi	Inhibit NE release
Beta-1	Heart, kidney	NE/Epi	↑ HR, ↑ contractility, ↑ renin
Beta-2	Lungs, GI, vessels	Epi > NE	Bronchodilation, vasodilation
Beta-3	Fat, bladder	NE/Epi	Lipolysis, relaxation

Organ Response Matrix EYE

- **Radial Muscle**
 - *Cholinergic (Parasympathetic) Response* – None
 - *Adrenergic Receptor Type* – alpha
 - *Noradrenergic (Sympathetic) Response* – Contraction causing Mydriasis
- **Sphincter Muscle**
 - *Cholinergic (Parasympathetic) Response* – Contraction causing Miosis
 - *Adrenergic Receptor Type* – None (—)
 - *Response* – None
- **Ciliary Muscle**
 - *Cholinergic (Parasympathetic) Response*

- Contraction for near vision
- *Adrenergic Receptor Type – beta*
- *Noradrenergic (Sympathetic) Response – Relaxation for far vision*

HEART

● **S-A Node**

- *Cholinergic (Parasympathetic) Response – Decreased heart rate*
- *Adrenergic Receptor Type – beta 1*
- *Noradrenergic (Sympathetic) Response – Increased heart rate*

● **Atria**

- *Cholinergic (Parasympathetic) Response – Decreased contractility*
- *Adrenergic Receptor Type – beta 1*
- *Noradrenergic (Sympathetic) Response – Increased contractility and conduction velocity*

● **A-V Node**

- *Cholinergic (Parasympathetic) Response – Decreased conduction velocity leading to A-V block*
- *Adrenergic Receptor Type – beta 1*
- *Noradrenergic (Sympathetic) Response – Increased conduction velocity*

● **Ventricles**

- *Cholinergic (Parasympathetic) Response – None*
- *Adrenergic Receptor Type – beta 2 (Note: While beta 1 dominates dynamically in ventricles, this accurately*

reflects your slide text)

- *Noradrenergic (Sympathetic) Response – Increased contractility and conduction velocity*

● **Arterioles (Skeletal/Visceral)**

- *Cholinergic (Parasympathetic) Response – Dilation*
- *Adrenergic Receptor Type – alpha, beta 2*
- *Noradrenergic (Sympathetic) Response – Constriction via alpha receptors, and Dilation via beta 2 receptors*

● **Arterioles (Skin/Salivary)**

- *Cholinergic (Parasympathetic) Response – None*
- *Adrenergic Receptor Type – alpha*
- *Noradrenergic (Sympathetic) Response – Constriction*

SYSTEMIC VEINS

- *Cholinergic (Parasympathetic) Response – None*
- *Adrenergic Receptor Type – alpha, beta 2*
- *Noradrenergic (Sympathetic) Response – Constriction via alpha receptors, and Dilation via beta 2 receptors*

LUNG

● **Bronchial Muscle**

- *Cholinergic (Parasympathetic) Response – Contraction leading to Bronchoconstriction*
- *Adrenergic Receptor Type – beta 2*
- *Noradrenergic (Sympathetic) Response – Relaxation leading to Bronchodilation*

- **Bronchial Glands**
 - *Cholinergic (Parasympathetic) Response* – Stimulation
 - *Adrenergic Receptor Type* – Unknown
 - *Noradrenergic (Sympathetic) Response* – Inhibition (?)

STOMACH

- **Motility**
 - *Cholinergic (Parasympathetic) Response* – Increased tone and movement
 - *Adrenergic Receptor Type* – alpha, beta 2
 - *Noradrenergic (Sympathetic) Response* – Decreased tone and movement
- **Sphincters**
 - *Cholinergic (Parasympathetic) Response* – Relaxation (usually)
 - *Adrenergic Receptor Type* – alpha
 - *Noradrenergic (Sympathetic) Response* – Contraction (usually)

URINARY BLADDER

- **Detrusor**
 - *Cholinergic (Parasympathetic) Response* – Contraction
 - *Adrenergic Receptor Type* – beta
 - *Noradrenergic (Sympathetic) Response* – Relaxation (usually)
- **Trigon/Sphincter**
 - *Cholinergic (Parasympathetic) Response* – Relaxation
 - *Adrenergic Receptor Type* – alpha

- *Noradrenergic (Sympathetic) Response* – Contraction

SKIN

- **Pilomotor Muscles**
 - *Cholinergic (Parasympathetic) Response* – None
 - *Adrenergic Receptor Type* – alpha
 - *Noradrenergic (Sympathetic) Response* – Contraction triggering hair erection
- **Sweat Glands**
 - *Cholinergic (Parasympathetic) Response* – Generalized secretion (Note: dynamically active via specialized Sympathetic Acetylcholine fibers)
 - *Adrenergic Receptor Type* – alpha
 - *Noradrenergic (Sympathetic) Response* – Slight localized secretion localized to palms

LIVER

- *Cholinergic (Parasympathetic) Response* – None
- *Adrenergic Receptor Type* – alpha, beta 2
- *Noradrenergic (Sympathetic) Response* – Glycogenolysis

PANCREAS

- **Islets**
 - *Cholinergic (Parasympathetic) Response* – Increased insulin and glucagon secretion
 - *Adrenergic Receptor Type* – alpha, beta 2
 - *Noradrenergic (Sympathetic) Response* – Decreased secretion via alpha receptors, and Increased secretion via beta 2 receptors

SALIVARY GLANDS

- *Cholinergic (Parasympathetic)*
Response – Profuse, watery secretion
- *Adrenergic Receptor Type* – alpha, beta 2
- *Noradrenergic (Sympathetic)*
Response – Thick, viscous secretion via alpha receptors, and Amylase secretion via beta 2 receptors

MALE SEX ORGANS

- *Cholinergic (Parasympathetic)*
Response – Erection
- *Adrenergic Receptor Type* – alpha
- *Noradrenergic (Sympathetic)*
Response – Ejaculation

CLINICAL CORRELATIONS

1. Horner's Syndrome

- Caused by the **interruption of the ipsilateral sympathetic pathway** tracking through the brainstem, upper thoracic spinal cord, cervical sympathetic chain, or the carotid plexus.
- Interruption **inactivates the pupillary dilator muscle and the superior tarsal muscle of the eyelid.**
- **Signs and Symptoms:**
 - Unilateral enophthalmos (**apparent backward displacement of the eyeball**).
 - **Ptoxis** (drooping of the upper eyelid).
 - **Miosis** (constricted pupil due to unopposed parasympathetic tone).
 - **Anhidrosis** (loss of sweating over the ipsilateral half of the face or forehead).

2. Raynaud's Disease

- Affecting **peripheral vascular**

innervation and tone.

- It primarily **targets the fingers, toes, edges of the ears, and tip of the nose.**

3. Causalgia

- A painful condition of the hands or feet caused by trauma or irritation to a major peripheral nerve trunk, most notably the **median nerve** or **sciatic nerve.**
- **Clinical Presentation:** Characterized by a **severe, persistent burning pain, accompanied by glossy skin changes, localized swelling, redness, altered sweating, and trophic changes to the nails.**

4. Hirschsprung's Disease (Congenital Megacolon)

- A congenital condition marked by **massive dilatation of a segment of the colon, accompanied by severe chronic constipation.**
- Caused by a congenital failure of parasympathetic terminal ganglia to develop within the wall of a large bowel segment. This leaves an **aganglionic, non-functional segment** that **blocks normal peristaltic movement**, causing massive backup and dilatation in the normal proximal colon.